



12573 - Planet Authentication in White-dwarf Systems (PAWS): Revealing the Demographics of Exoplanets Orbiting White Dwarfs

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		MIRI Imaging	(1) CPD-69-177
	2		MIRI Imaging	(2) UPM-J0812-3529
	3		MIRI Imaging	(3) GD-140

ABSTRACT

Understanding how post-main-sequence stellar evolution affects exoplanet populations is a key question in the field today, but so far, relatively few planets have been confidently detected around white dwarfs. Recent JWST observations have detected numerous planet candidates in white dwarf systems, but these discoveries require additional observations before they can be confirmed. We request MIRI imaging observations of an unbiased,

JWST Proposal 12573 (Created: Monday, March 23, 2026, 1:00:43PM Eastern Standard Time) - Overview

volume-limited sample of 16 nearby white dwarfs with directly imaged planet candidates from Cycle 2 JWST observations. Our additional MIRI observations will let us measure each candidate's proper motion and distinguish real planets from background contaminants. Our program will result in the confirmation of numerous giant planet candidates (likely including some of the coldest exoplanets ever imaged) and a first measurement of the occurrence rate of giant planets around white dwarfs. Based on main-sequence planet occurrence rates, we expect to confirm 5 ± 2 planets in our sample (with 99.6% probability of detecting at least one planet). If our actual number of detected planets differs from this estimate significantly, it would suggest that our understanding of post-main-sequence planetary evolution is incomplete.

OBSERVING DESCRIPTION

We will collect MIRI imaging for 16 white dwarf systems using the same filters and exposure times as in the first epoch: F2100W: 12 min, F1800W: 5 min, and F770W: 1 min. A 4-point cycling dither pattern will be employed to mitigate bad pixels and enable background correction, consistent with the first epoch of observations. This consistent approach ensures the detection of the same sources at the same SNR that was achieved in the first epoch and gives multiple independent checks (in each filter) on the proper motion measurements. The charged time for this program, including overheads, is 24.1 hours.

Due to the proximity of the white dwarf systems in the sample, all exhibit high or very high proper motion (up to 3.6 arcsec/year). The white dwarfs in this sample will have shifted on the detector by 2-65 pixels between the two epochs. Using PSF fitting, we are capable of measuring the positions of the faintest exoplanet targets within the sample to <0.4 pixels. Therefore, with a two-year baseline between the epochs we can achieved >5 sigma detection of proper motion even for the faintest candidates in the field of view.

Since there are currently no high-precision distortion maps for the MIRI 18 and 21 micron imagery, we will constrain the PA of the second epoch of observation to match the first (± 1 degree). This ensures that the lack of a high-fidelity distortion correction does not compromise our ability to conduct high-precision proper motion measurements. Additionally, keeping the PA consistent ensures that we capture the same candidates at wide angular separations near the edge of the field of view, which might otherwise be cut off the detector if the field rotation differs from the first epoch.

Proposal 12573 - Targets - Planet Authentication in White-dwarf Systems (PAWS): Revealing the Demographics of Exoplanets Orbitin...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	CPD-69-177	RA: 03 10 31.0196 (47.6292483d) Dec: -68 36 3.38 (-68.60094d) Equinox: J2000	Proper Motion RA: 39.668 mas/yr Proper Motion Dec: -103.23700003027625 mas/yr Parallax: 0.0961834" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[White dwarfs]				
	(2)	UPM-J0812-3529	RA: 08 12 27.0660 (123.1127750d) Dec: -35 29 43.32 (-35.49537d) Equinox: J2000	Proper Motion RA: -65.479 mas/yr Proper Motion Dec: -29.204000088611792 mas/yr Parallax: 0.0895064" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[White dwarfs]				
	(3)	GD-140	RA: 11 37 5.1041 (174.2712671d) Dec: +29 47 58.29 (29.79953d) Equinox: J2000	Proper Motion RA: -147.714 mas/yr Proper Motion Dec: -12.532999994618876 mas/yr Parallax: 0.0637329" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[White dwarfs]				

Proposal 12573 - Observation 1 - Planet Authentication in White-dwarf Systems (PAWS): Revealing the Demographics of Exoplanets ...

Observation	Proposal 12573, Observation 1										Mon Mar 23 18:00:43 GMT 2026
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	CPD-69-177	RA: 03 10 31.0196 (47.6292483d) Dec: -68 36 3.38 (-68.60094d) Equinox: J2000			Proper Motion RA: 39.668 mas/yr Proper Motion Dec: -103.23700003027625 mas/yr Parallax: 0.0961834" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[White dwarfs]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	2	F1800W	FASTR1	12	2	1	Dither 1	4	8	277.504	
	3	F2100W	FASTR1	12	5	1	Dither 1	4	20	710.41	
Special Requirements	Aperture PA Range 310.33544897 to 312.33544897 Degrees (V3 305.5 to 307.5)										

Proposal 12573 - Observation 2 - Planet Authentication in White-dwarf Systems (PAWS): Revealing the Demographics of Exoplanets ...

Observation	Proposal 12573, Observation 2										Mon Mar 23 18:00:43 GMT 2026
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	UPM-J0812-3529	RA: 08 12 27.0660 (123.1127750d) Dec: -35 29 43.32 (-35.49537d) Equinox: J2000			Proper Motion RA: -65.479 mas/yr Proper Motion Dec: -29.204000088611792 mas/yr Parallax: 0.0895064" Epoch of Position: 2000					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
Category=Star											
Description=[White dwarfs]											
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	2	F1800W	FASTR1	12	2	1	Dither 1	4	8	277.504	
	3	F2100W	FASTR1	12	5	1	Dither 1	4	20	710.41	
Special Requirements	Aperture PA Range 63.53544897 to 65.53544897 Degrees (V3 58.7 to 60.7)										

Proposal 12573 - Observation 3 - Planet Authentication in White-dwarf Systems (PAWS): Revealing the Demographics of Exoplanets ...

Observation	Proposal 12573, Observation 3										Mon Mar 23 18:00:43 GMT 2026
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	GD-140	RA: 11 37 5.1041 (174.2712671d) Dec: +29 47 58.29 (29.79953d) Equinox: J2000			Proper Motion RA: -147.714 mas/yr Proper Motion Dec: -12.532999994618876 mas/yr Parallax: 0.0637329" Epoch of Position: 2000					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
Category=Star											
Description=[White dwarfs]											
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	2	F1800W	FASTR1	12	2	1	Dither 1	4	8	277.504	
	3	F2100W	FASTR1	12	5	1	Dither 1	4	20	710.41	
Special Requirements	Aperture PA Range 115.83544897 to 117.83544897 Degrees (V3 111.0 to 113.0)										